

John Holik

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Software engineer specializing in applied machine learning and LLM application development. Second author on a manuscript under review at *Automation in Construction* (Elsevier); shipped a live retrieval-augmented chatbot (ElectriAI, on Cloudflare) and a multi-service LLM web application (CapSight AI). Strengths in model evaluation against human ground truth and reproducible experimentation.

EDUCATION

Florida Gulf Coast University, Fort Myers, FL

May 2026

B.S. in Computer Software Engineering · Dean's List (Junior and Senior Years)

Relevant coursework: Data Structures and Algorithms, Machine Learning, Data Mining, Data Engineering, Operating Systems, Computer Security, Software Engineering, AI Entrepreneurship

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash, PowerShell

ML and Data: scikit-learn, pandas, NumPy, SciPy, model evaluation (ROC-AUC, precision/recall/F1, calibration, Cohen's kappa), uncertainty quantification, clustering (K-means, UMAP, HDBSCAN)

AI and LLM: OpenAI, Anthropic (Claude), and Gemini APIs, prompt engineering, structured JSON-schema output, LLM evaluation vs human ground truth, custom agent and Claude Code skill development, RAG and embeddings (deployed retrieval-augmented chatbot)

Web and Backend: React, FastAPI, Node.js/Express, PostgreSQL, Cloudflare Pages/Workers (deployed)

Tools: Git/GitHub, Jupyter, matplotlib/plotly, LaTeX, Linux

EXPERIENCE

Undergraduate Research Assistant, then Independent Researcher (ElectriAI)

Aug 2025 – Present

U.A. Whitaker College of Engineering, Florida Gulf Coast University

- Built an end-to-end LLM pipeline over a dataset of electrical-construction YouTube videos and comment threads: collection, schema-constrained GPT classification, human annotation, and rigorous evaluation. Second author on the resulting manuscript, "Identifying Knowledge Bottlenecks in Electrical Construction from Practitioner Discussion on YouTube Using Large Language Models," submitted to *Automation in Construction* (Elsevier) July 2026.
- Processed 794 video transcripts and 66,899 comment threads into 16,862 structured question-and-answer records under a 76-theme, ten-category taxonomy; classified 12,933 questions into ten functional forms.
- Validated model labels against an independent 26-annotator human consensus: 0.847 Cohen's kappa, 86.6% accuracy, 0.858 weighted F1, with per-class precision, recall, and F1.
- Built a reproducible evaluation harness (fixed seeds, versioned data) and released the supporting data and code as a citable Zenodo archive.
- Built and deployed the public ElectriAI companion web app (React, Tailwind) on Cloudflare Pages, and generated its chatbot knowledge base from the taxonomy rather than hand-authoring it: 288 pages (263 question-family, 11 question-type, 10 answer-type, 3 analytics, 1 overview), each with computed answer-rate and per-year statistics and verbatim threads carrying (V:videoID)/(Q:commentID) provenance. Chunked under a 6,000-character cap, splitting long pages at H2 boundaries with the page title prefixed to each chunk for context: 297 chunks, embedded offline via `gemini-embedding-001` at 768 dimensions (RETRIEVAL_DOCUMENT) and shipped as static JSON, so no vector database is required.
- Built the retrieval-to-answer path: cosine similarity over the 297 vectors plus a keyword-overlap boost and regex intent routing to the analytics pages, capped at two chunks per page for diversity, top eight retained. The prompt renders the top four in full and the next four as 320-character snippets, each tagged with its score, and constrains the model to those passages, verbatim citation tokens, and no number absent from them; a post-generation regex check strips ungrounded numbers and retries. Streams from `gemini-3.6-flash` through a Cloudflare Worker proxy (API-key isolation, KV rate limiting). Live at youtube.electriai.com.

PROJECTS

CapSight AI, LLM Pitch-Deck Grading Platform

React, FastAPI, PostgreSQL, Node/TypeScript, OpenAI

- Built a three-service application that scores uploaded pitch decks across six categories with an LLM, persisted to PostgreSQL with a history view.
- Enforced structured output via OpenAI `json_schema` and made grading deterministic (seed 42, temperature 0) for repeatable results.
- Hardened the grading core with adaptive retry, score clamping, an unreadable-deck sentinel, a fallback result, and non-blocking database writes so grading still returns if the database is down.

Credit-Risk ML, Loan-Default Prediction

Python, scikit-learn

- Benchmarked 14 models on 32,416 loans with an explicit train-vs-test overfit check; selected Gradient Boosting at 0.95 test ROC-AUC with a small generalization gap.
- Added probability calibration (Brier 0.05), ROC/PR curves, partial dependence, and permutation importance; fully reproducible (seed 42, pinned versions).

Red Tide Reanalysis, Senior Capstone

Python, NumPy, scikit-learn, Jupyter Book

- Implemented and compared four ensemble uncertainty-quantification methods (Bootstrap, GLUE, EnKF, LPU) feeding a Random Forest bloom classifier; published as a reproducible Jupyter Book.

Additional: UFC relational database (five-table normalized PostgreSQL schema); Selenium/TestNG end-to-end UI suite (eight areas, Java/Maven); Pro-Code AI Socratic programming tutor (FastAPI, PostgreSQL, React).